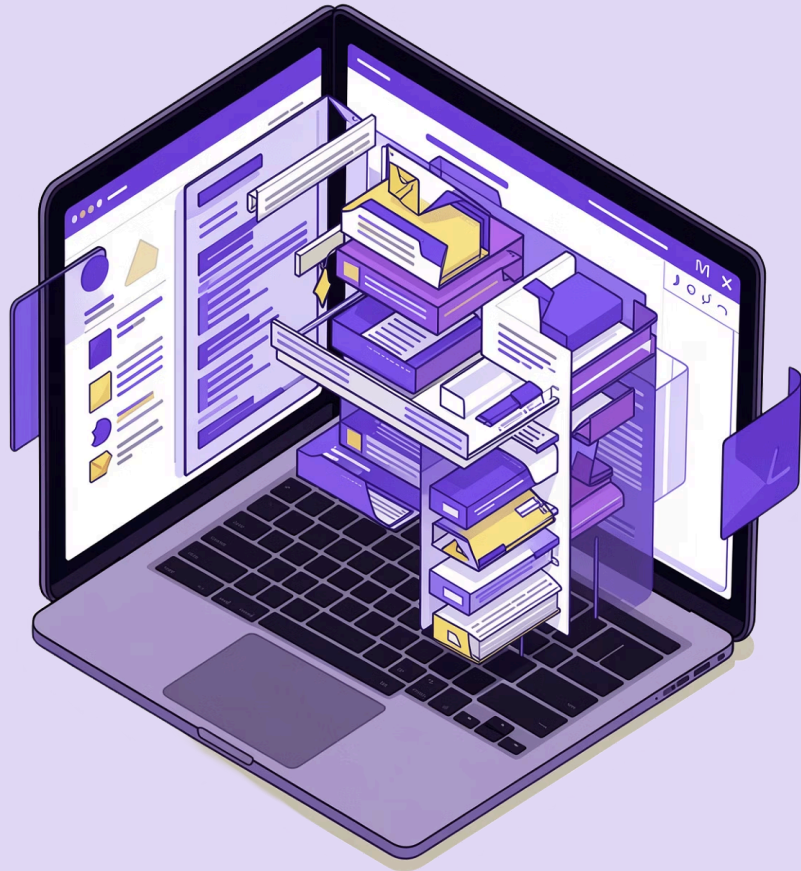




InboxToTask: AI-Powered Email-to-Action System

IS 492 Checkpoint 1 | Team Name: InboxToTask |
<https://github.com/saniashah012/InboxToTask/>

Matthew Zheng - Product Strategy and Research

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Hua Tong - UX/UI Design and User Research

Sania Shah - Product ideation/design and research

The Problem: Email Overload Kills Productivity

Buried Commitments

Emails contain requests, deadlines, and commitments—but users manually transfer them to calendars and task managers, one by one.

Growing Inbox, Growing Stress

Expanding inboxes bury important actions under layers of noise, leaving unresolved commitments invisible.

Documented Harm

Kern et al. (2024) shows heavy email loads drive measurable emotional irritation and sustained workflow stress.



What the Research Tells Us

Academic insights informed our approach to transform email chaos into clear action.

Email Overload & Cognitive Strain

Heavy email loads cause measurable stress and workflow disruption. InboxToTask mitigates this by extracting key actions, reducing the mental burden of sifting through noise.

(Kern et al. 2024)

Commitment Detection Gaps

Traditional ML approaches often fail to reliably detect commitments across diverse writing styles and domains. Our solution employs advanced semantic understanding to overcome these limitations.

(Azarbonyad et al. 2019)

Natural Language Ambiguity

Human language is nuanced, requiring context and implicit meaning to be fully understood. InboxToTask leverages sophisticated AI to interpret vague requests into actionable tasks.

Integration Gap

Existing tools, like email clients and general AI assistants, don't seamlessly close the loop from an email to a tracked action. InboxToTask bridges this critical gap for productivity.

Who We're Building For & What They Need

InboxToTask is designed to empower professionals by directly addressing the challenges of email management and task prioritization.

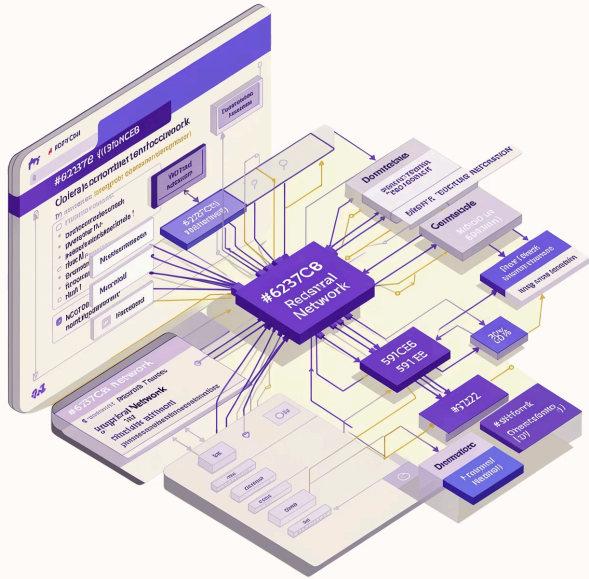
Target Audience

- Knowledge workers drowning in email (managers, coordinators, professionals with high email volume)
- People juggling multiple projects and deadlines
- Teams that rely on email as their primary communication channel

Key Tasks & Desired Outcomes

- Automatically surface actionable items from email noise
- Reduce time spent manually organizing tasks (from hours/week to minutes)
- Never miss a deadline or commitment again
- Reclaim focus time by eliminating email-to-task friction

Why This Needs AI



Language Is Ambiguous by Nature

"Can you send slides Thursday?" vs. *"I'll send slides Thursday"* — same words, opposite actions required. No rule-based system can reliably distinguish them.

Context Is Everything

References, implicit timing cues, and conversation thread history all shape the true meaning of an email. Meaning lives between the lines.

GenAI Bridges the Gap

Generative AI interprets natural language nuance and transforms messy, unstructured email content into clean, structured next steps.

Existing Tools Fall Short

Traditional Email Clients

Offer manual tools (flags, folders) but don't automatically identify which messages contain actionable commitments—users must manually flag and organize.

AI Assistants (Copilot, Claude)

Require explicit user prompting each time. Output stays siloed in chat—never integrated with task managers or calendars.

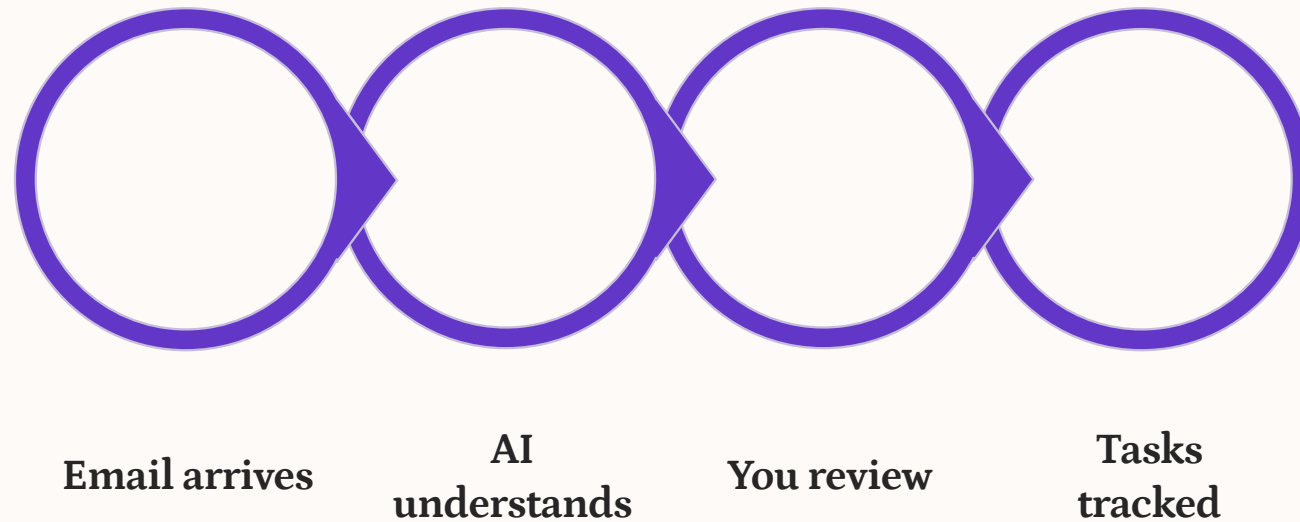
ML Commitment Detection

Azarbonyad et al. (2019) showed ML approaches degrade significantly across diverse writing styles and domains—too brittle for real-world inboxes.

 No existing tool closes the loop from inbox to action automatically.

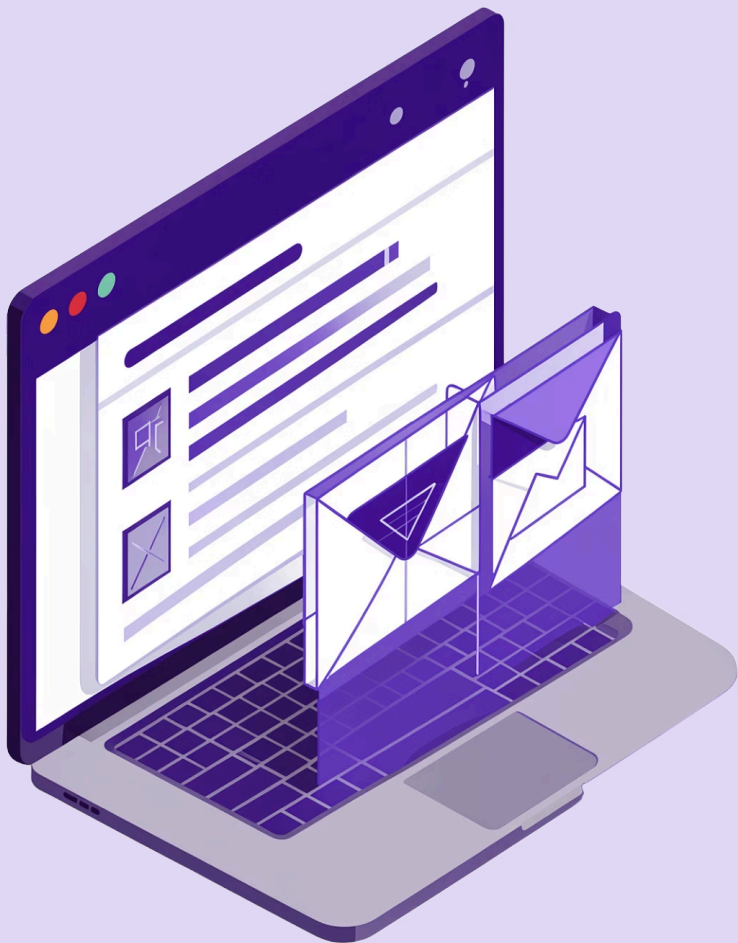
Our Solution: InboxToTask

Built for busy professionals who want to reclaim time from email management and turn inbox chaos into clear next steps.



How It Works

- Email arrives with requests and commitments
- AI analyzes and suggests actionable next steps
- You review and approve before syncing
- Tasks automatically tracked in your calendar and task manager



Why GenAI Is Essential



Semantic Reasoning

Distinguishes task-creating requests from passive commitments to track — a distinction that requires genuine language understanding, not pattern matching.



Context Awareness

Resolves pronouns, implicit references, and email-thread relationships to surface the true intent behind any message.



Flexibility

Handles varied writing styles, tones, and implicit information without rigid rules — adapting to every sender's habits naturally.

Checkpoint 2: Validation & Next Steps

Core Validation

Test prompt logic on a diverse corpus of sample emails spanning requests, commitments, hard deadlines, and deliberately ambiguous cases.

Key Risks

- AI hallucinations producing false task extractions
- Anchoring bias from early prompt framing
- Privacy and PII exposure in email content

Mitigations

- **Pre-commitment prompting** to reduce hallucination risk
- **Source linking** so every suggestion traces back to its email
- **Client-side PII scrubbing** before any data leaves the device
- **Zero-data retention agreements** with API providers